

B.2.3 Exam Practice 3: Hardware

Date: 4/27/2026, 2:08:15 PM

Time Spent: 11:01

Score: 80%

Passing Score: 80%



Question 1

✓ Correct

What can a technician use on a 20-pin main power connector for the motherboard? (Select two.)

- ☐ 4-pin x5
- ☒ 24-pin ✓ Correct
- ☐ 8-pin x2 and 4-pin x1
- ☒ 20-pin ✓ Correct

Explanation

The technician can use a 20-pin main power adapter for the motherboard. This is the adapter that was specifically made for it.

A 24-pin adapter supplies power to the motherboard but can connect to a 20-pin. In most cases, a 24-pin adapter has 4 pins that are detachable. This is not always 100% the case.

4-pin x5 adapters would not work with the 20-pin main power connector. 4-pin connectors typically connect to case fans and other accessories devices.

A technician cannot use 8-pin x2 and 4-pin x 1. 8-pin for all multi-core processors use this connector since some power supplies have two 4-pin connectors that they can use side by side.

Related Content

 3.1.4 Power Supply Connectors

 3.1.5 20-pin to 24-pin Motherboard Adapter

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Question 2

✔ Correct

Regarding a personal computer, what houses internal components or provides input/output functions or external storage? (Select two.)

☒ Peripheral device ✔ Correct

☒ System case ✔ Correct

☐ USB

☐ HDMI

Explanation

Peripheral devices typically perform the input function (keyboard, mouse, microphone, and camera), output (monitor and speakers), or external storage.

The System case houses the internal components, including the motherboards, central processing unit (CPU), system memory modules, adapter cards, fixed disks, and power supply unit.

Universal Serial Bus (USB) is the standard means of connecting most peripheral device types to a computer. A host controller manages the USB.

A high-definition multimedia interface (HDMI) is the most widely used video interface. HDMI supports both video and audio. HDMI rates as either standard or high speed.

Related Content

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Question 3

✔ Correct

A remote user contacts the IT desk to report that they accidentally dropped a mobile device. It is no longer responsive when trying to tap on icons. Which component is likely damaged and is causing the problem?

☒ Digitizer ✔ Correct

☐ Tempered glass

☐ Display panel

☐ Protective glass

Explanation

A touch screen can also be referred to as a digitizer due to the way it converts analog touch input to digital software instructions. The digitizer is sandwiched between a layer of protective glass and the display panel.

A display panel is only a single layer within the screen assembly for a mobile device. A display sits below the digitizer.

Protective glass covers the digitizer on a mobile device. The protective glass and digitizer usually adhere together.

The touchscreen itself is covered by a thin layer of scratch-resistant, shock-resistant tempered glass, such as Corning's Gorilla Glass.

Related Content

2.1.9 Display Components

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Question 4

✔ Correct

Which of the following causes the ink in an inkjet printer to break up into small pumps, which give an electrical charge when they leave the print head and can either stick to the paper or fall off?

- ☐ Thermal method
- ☒ Epson design ✓ Correct
- ☐ Transfer roller
- ☐ Imaging drum

Explanation

Epson design, the nozzle contains a piezoelectric element, which changes shape when a voltage is applied. The steady stream is the ink sent through the print head, and the ink that falls off is recycled.

The thermal method is when the ink at each nozzle in the print head is heated, creating a bubble. When the bubble bursts, it sprays ink through the nozzle and draws more ink from the reservoir.

The imaging drum is a round rotating cylinder conditioned by the primary charge roller (PCR) in a laser printer.

The transfer roller applies a positive charge to the underside of the paper in a laser printer.

Related Content

 10.1.11 Printer Security

 10.2.3 Inkjet Printer Imaging Process

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Question 5

✔ Correct

Which universal serial bus (USB) 2 connector is square with a beveled top that a technician can use for printers?

- ☐ Type B Micro
- ☐ Type A
- ☒ Type B ✓ Correct
- ☐ Type B Mini

Explanation

Type B connects to large devices. The connector and port are square, with a beveled top.

Type B Mini is a smaller peripheral device connector. This connector was seen on early digital cameras but is no longer widely used.

Type B Micro is an updated connector for smaller devices such as smartphones and tablets. The micro connector is distinctly flatter than the older mini type of connector.

Type A connects to the host and some types of peripheral devices. The connector and port are flat rectangle-shaped. A technician should insert the connector with the USB symbol facing up.

Related Content

 2.1.4 Universal Serial Bus Cables

 2.1.5 USB Standards

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Question 6

✔ Correct

An employee uses a device that gives them authentication to release the print job. What is this device?

- ☐ Print server
- ☐ Audit log
- ☐ User authentication
- ☒ Badging ✓ Correct

Explanation

Badging means the print device has a smart card reader. An employee must present an ID badge to the reader to start the print job.

A print server is a hardware and firmware that allows client computers to connect directly over the network without going via a server computer.

An audit log can identify printed documents that have gone missing or identify the unauthorized release of information.

A company can configure user authentication on a print share. User authentication means that the printer-sharing server or print device will only accept print jobs from authorized user accounts.

Related Content

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Question 7

✕ Incorrect

The TIA/EIA-568-C Commercial Building Telecommunications Cabling Standards have specific recommendations for healthcare facilities with Power over Ethernet (PoE) 802.3bt installations, and for running distribution system cables to wireless access points. Which Cat does this refer to?

- ☐ Cat 5e
- ☐ Cat 5
- ☒ Cat 6 ✕ Incorrect
- ☐ Cat 6A ✓ Correct

Explanation

Cat 6A is recommended for healthcare facilities since it supports 10 Gbps over 100m, but the cable is bulkier and heavier than Cat 5e and Cat 6. The installation requirements are more stringent.

Cat 5 cable supports the older 100 Mbps fast Ethernet standard. It is no longer commercially available, so a technician will need to rewire the network cable with Cat 5 to support Gigabit Ethernet.

Cat 5e cable supports Gigabit Ethernet links for client computers, but can now opt to install Cat 6 cable.

Cat 6 cable is more reliable than Cat 5e and supports 10 Gbps but over a reduced range.

Related Content[5.3.4 Cat Standards](#)[resources\questions\pt_core1_220_1201_3_2_012.question.xml](#)

Question 8

 Correct

An IT technician installs a solid-state drive (SSD) into a motherboard connector that enables high-performance storage while conserving space inside the system case. What has the IT technician installed?

- ☐ SATA
- ☒ M.2 ✓ Correct
- ☐ eSATA
- ☐ PCI

Explanation

M.2 adapters can be of different lengths. The adapter card is inserted at an angle, and then pushed into place and secured with a screw.

Serial advanced technology attachment (SATA) ports connect to one or more fixed drives. Technicians can also use SATA to connect removable drives, such as tape and optical drives.

External SATA (eSATA) is a standard for attaching external drives with a 2 m (78 in.) cable.

A peripheral component interconnect (PCI) is a legacy bus type superseded by PCI Express. PCI ports can be included on a PCIe motherboard to support legacy adapter cards, but PCI cards cannot fit into PCIe slots.

Related Content

-  2.2.4 Motherboard Storage Connectors
-  3.2.3 Solid-State Drives
-  3.2.11 Optical Drives
-  4.2.13 Troubleshooting Problems with Storage Drives & RAID Arrays
-  8.2.7 Cloud File Storage
-  12.4.2 Storage
-  15.2.2 Windows Domains and Active Directory
-  21.1.4 Backup Media Requirements

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Question 9

✔ Correct

A traveling salesperson receives an old laptop to use while a new one is on back-order. The salesperson complains that the laptop's visuals and brightness are horrible. An IT technician states that the fluorescent backlight is failing. What powers the backlight?

- ☐ IPS
- ☒ Inverter ✓ Correct
- ☐ TFT
- ☐ Digitizer

Explanation

An inverter component is used to convert from the DC power supplied by the motherboard to the AC power for the bulb. Older laptops might use an inverter component to power a fluorescent backlight.

A touch screen can also be referred to as a digitizer because of the way it converts analog touch input to digital software instructions.

In-plane switching (IPS) is a display technology that uses crystals that rotate rather than twist. The main benefit is to deliver better color reproduction at a wider range of viewing angles.

In the types of flat panels used for computer and mobile device displays, the liquid crystal elements and transistors are placed on a thin film transistor (TFT).

Related Content

2.1.9 Display Components

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Question 10

✔ Correct

An iPhone user needs to charge their phone and notices that it only uses a certain type of connector. What type of connector does the iPhone use?

- ☐ USB
- ☐ Thunderbolt
- ☐ SATA
- ☒ Lightning port ✔ Correct

Explanation

A Lightning port is found only on Apple's mobile devices. The Lightning connector is reversible.

Universal Serial Bus (USB) is the standard means of connecting most peripheral device types to a computer. A host controller manages the USB.

Thunderbolt displays interfaces like the DisplayPort or HDMI and a general peripheral interface like USB.

SATA (serial advanced technology attachment) is the standard means of connecting internal storage drives within a desktop PC. SATA uses cables of up to 1 m terminated with compact 7-pin connectors. Each SATA host adapter port supports a single device.

Related Content

2.1.12 Lightning Interface

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Question 11

✕ Incorrect

A system administrator implemented a physical device that expedites cryptographic operations and manages cryptographic keys. What is this device the system administrator implemented?

☐ Boot options

☒ TPM ✕ Incorrect

☐ Secure boot

☐ HSM ✓ Correct

Explanation

A hardware security module (HSM) is a secure USB key or thumb drive used to store cryptographic material.

A trusted platform module (TPM) is a specification for hardware-based storage of digital certificates, cryptographic keys, and hashed passwords.

Boot options define how the system firmware searches devices for a boot manager. It is one of the most important parameters in a system setup.

Secure boot is a unified extensible firmware interface (UEFI) feature designed to prevent a computer from being hijacked by malware. Under secure boot, the computer firmware configuration has cryptographic keys that can identify trusted code.

Related Content

4.1.8 Trusted Platform Modules

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Question 12

✕ Incorrect

A technician researches flat-screen displays to find the perfect model for a conference room. The screen should feature a high viewing angle as the conference room is wide. Which technology does the technician look for when purchasing a flat-screen display?

☐ VA☐ IPS ✓ Correct☐ LED☒ OLED ✕ Incorrect**Explanation**

In-plane switching (IPS) uses crystals that rotate rather than twist. The main benefit is to deliver better color reproduction at a wider range of viewing angles. Most IPS panels support 178/178 degree horizontal and vertical viewing angles.

Vertical alignment (VA) uses crystals that tilt rather than twist or rotate. This technology supports a wide color gamut and the best contrast-ratio performance.

An array of light-emitting diodes (LEDs) can provide illumination in a flat panel. Most smartphone and tablet screens use edge lighting where the LEDs are at the top or bottom of the screen.

A separate LED generates each pixel in an organic LED (OLED) display, or technically, an advanced matrix OLED (AMOLED) display.

Related Content[2.1.8 Display Types](#)[resources\questions\pt_core1_220_1201_3_1_001.question.xml](#)

Question 13

✔ Correct

A manager uses a dot matrix printer that uses specific types of paper. Which of the following papers does the printer use? (Select two.)

- ☐ Thermal paper
- ☒ Tractor-fed paper ✔ Correct
- ☒ Impact paper ✔ Correct
- ☐ Replaceable ribbon

Explanation

Impact paper makes multiple copies of a document in the same pass. A sheet of impact paper is between each sheet of plain paper, and when the print head strikes, the same mark remains on each sheet.

Tractor-fed paper has removable, perforated side strips. The holes in these strips secure over studded rollers at each end of the platen.

The thermal paper contains chemicals designed to react and change color as it is heated by the heating element within the printer to create images.

Impact printers use a replaceable ribbon with a cartridge device that slots over or around the carriage of the print head.

Related Content

10.2.6 Impact Printer Maintenance

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Question 14

✔ Correct

A technician uses an adapter card that supplies power over the bus, resulting in no need for a separate power cable. Because this adapter can vary in different widths and lengths, the technician should check to ensure that the adapter fits on the motherboard. What type of adapter has these specifications?

- ☐ SSD
- ☒ M.2 ✓ Correct
- ☐ mSATA
- ☐ HDD

Explanation

The M.2 adapter card form factor is considerably smaller than a PCIe adapter. It is oriented horizontally rather than vertically, so the interface is often used on laptops and PC motherboards.

A solid-state drive (SSD) uses flash memory technology to implement persistent mass storage. Flash memory performs much better than the mechanical components used in hard drives.

A hard disk drive (HDD) stores data on metal or glass platters coated with a magnetic substance. The top and bottom of each platter are accessed by its own read/write head, moved by an actuator mechanism.

A mini-serial ATA (mSATA) allows a technician to plug an SSD packaged as an adapter card into the motherboard's combined data and power port.

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 15.2.2 Windows Domains and Active Directory

 21.1.4 Backup Media Requirements

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Question 15

✓ Correct

A technician notices that the printer's output was not expected and needs to check the printer's control panel to investigate the problem. What does the technician need to check?

☐ Developer roller

☐ Toner

☒ Calibration ✓ Correct

☐ Maintenance kit

Explanation

Calibration is how the printer determines the appropriate print density or color balance. Most printers calibrate themselves automatically.

A maintenance kit is a replacement feed roller, transfer roller, and fuser unit. The printer's internal copy count of the number of printed pages guides the replacement kit.

Toner is composed of a fine compound of dyestuff and wax or plastic particles. The toner carries the same negative charge polarity as the drum.

The developer roller is very close to the photosensitive drum. The toner is fed evenly onto a magnetized developed roller from a hopper.

Related Content

 10.2.1 Laser Printer Imaging Process

 10.2.2 Laser Printer Maintenance

 10.2.4 Inkjet Printer Maintenance

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Question 16

✔ Correct

What component can be used in a camera to hold digital image storage?

- ☐ Optical drive
- ☐ Flash drive
- ☒ Memory card ✔ Correct
- ☐ HDD

Explanation

Memory card is a form factor used in consumer digital imaging products, such as digital still and video cameras, and to expand smartphone and tablet storage.

Flash drives are flash memory boards with a USB connector and protective cover.

Optical drives rate according to their data transfer speed. An optical drive can perform recording/rewriting and markets with three speeds.

A hard disk drive (HDD) stores data on metal or glass platters coated with a magnetic substance. The top and bottom of each platter are accessed by its own read/write head, moved by an actuator mechanism.

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Question 17

✔ Correct

What channel system requires support from the CPU, memory controller, and motherboard but not from the RAM devices?

- ☐ ECC RAM
- ☐ Triple-channel
- ☐ Single-channel
- ☒ Dual-channel ✔ Correct

Explanation

Dual-channel memory controllers effectively have two 64-bit pathways through the bus to the CPU, meaning that they can send 128 bits of data per transfer rather than 64 bits.

Single-channel memory means that there is one 64-bit data bus between the CPU, memory controller, and RAM devices.

Technicians can use triple-channel or quad-channel memory controllers in some CPUs and supporting chipsets. If the technician does not install the full complement of modules, the system will revert to as many channels as are populated.

Technicians use error correction code (ECC) RAM for workstations and servers that require high reliability.

Related Content

3.3.5 Multi-channel System Memory

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Question 18

✔ Correct

Which of the following allows reducing the number of cables to the minimum required to minimize clutter within the chassis and improve airflow and cooling?

- ☐ Redundant power supply
- ☐ Thermal pad
- ☐ Heat sink
- ☒ Modular power supply ✔ Correct

Explanation

A modular power supply has power connector cables that are detachable from the unit.

A redundant power supply requires a compatible motherboard. This configuration is more common in server systems than on desktop PCs. Typically, each PSU plugs into a backplane on a server and is hot-swappable.

A heat sink is a block of copper aluminum with fins. The fins expose a larger surface area to the air around the component to achieve a cooling effect by convection.

A thermal pad is a solid compound at room temperature but softens when heated.

Related Content

3.1.6 Modular Power Supplies

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Question 19

✔ Correct

What tool is also known as a universal serial bus (USB) drive, thumb drive, or pen drive?

- ☐ HDD
- ☐ Optical drive
- ☒ Flash drive ✓ Correct
- ☐ Memory card

Explanation

Flash drives are flash memory boards with a USB connector and protective cover.

Optical drives rate according to their data transfer speed. An optical drive can perform recording/rewriting and markets with three speeds.

Memory card is a form factor used in consumer digital imaging products, such as digital still and video cameras, and to expand smartphone and tablet storage.

A hard disk drive (HDD) stores data on metal or glass platters coated with a magnetic substance. The top and bottom of each platter are accessed by its own read/write head, moved by an actuator mechanism.

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Question 20

✕ Incorrect

What universal serial bus (USB) does a technician use that allows a device to send and receive simultaneously?

☒ USB Type A ✕ Incorrect

☐ USB 3.0 ✓ Correct

☐ USB 2.0

☐ USB Type B

Explanation

USB 3.0 is a new type of connector form factor and has upgraded transfer rates which are full-duplex.

USB 2.0 is a high-speed standard that specifies a data rate of 480 Mbps shared between all devices attached to the same host controller. The USB 2.0 is half-duplex, which means that each device can send or receive, but not simultaneously.

USB 2 Type B connects to large devices. The connector and port are square, with a beveled top.

USB 2 Type A connects to the host and some types of peripheral devices. The connector and port are flat rectangle-shaped. A technician should insert the connector with the USB symbol facing up.

Related Content

2.1.4 Universal Serial Bus Cables



2.1.5 USB Standards



3.2.3 Solid-State Drives

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